

PAPER_1198_UPDATE_Multi_Derivation_Session

Daniel T. Murphy

PAPER_1198_UPDATE — Multi-Derivation Session Extensions

Original: PAPER_1198 ρ_{vac} SCm Derivation & Particle Physics Proof Set

Update: Session June 2026

Status: Update

Session Extensions

Closed forms confirmed in calculator:

- $\Lambda_{\text{QCD}} = D_{\text{crit}}^2/\pi = \mathbf{215.2 \text{ MeV}}$ (1.29% vs 218)
- $\alpha_s(M_Z) = 0.1179$ (anchored)
- V_{us} CKM, $\Delta m^2_{21/31}$ neutrino mass splittings
- See PAPER_1219 Reading B for Riemann projection
- See PAPER_1217 for mass ratios